Fault Code: 198 (Aftermarket and OEM)

Indicator Lamp Circuit - Voltage Above Normal or Shorted to High Source

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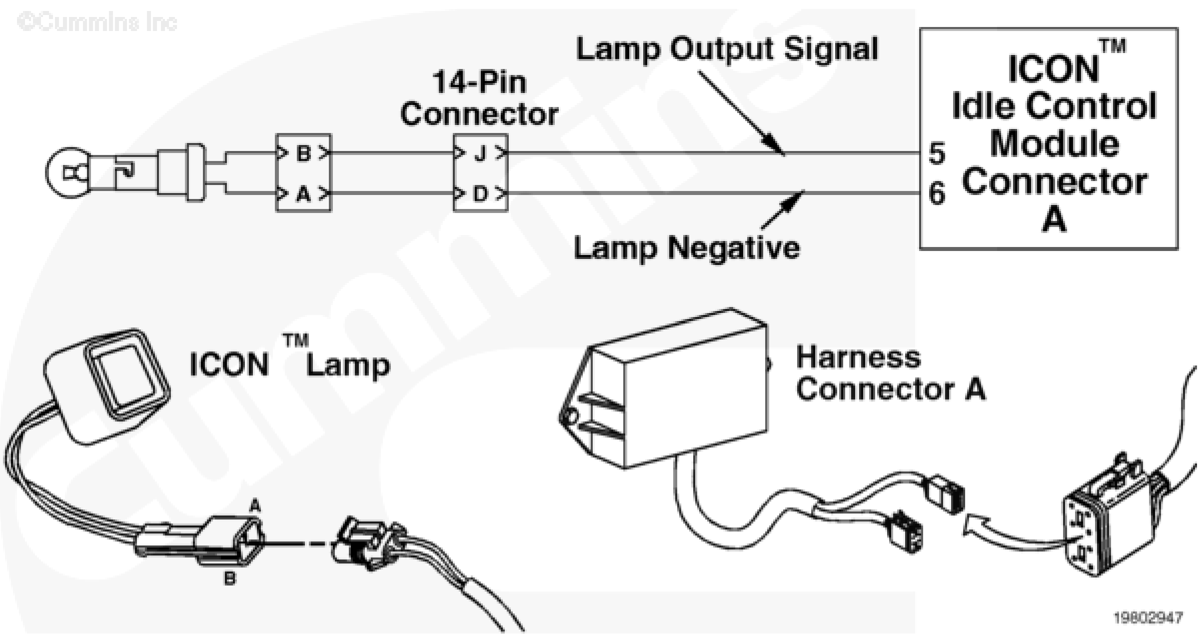
Overview

Codes	Reason	Effect
Fault Code: 198 PID(P): SPN: FMI: Lamp: SRT:	Indicator Lamp Circuit - Voltage Above Normal or Shorted to High Source. High voltage detected at the ICON™ lamp or LED circuit when low voltage was expected by the ICON™ idle control module.	The ICON™ system will be disabled. **Only** mandatory shutdown will be enabled.



Circuit Description

The ICON™ lamp or LED circuit turns on the ICON™ lamp to indicate when the ICON™ system is active. In addition, ICON™ fault codes will be flashed out on this lamp. The lamp or LED circuit requires a specific flash timing (on/off timing). If the on/off voltage is incorrect, the ICON™ system will be disabled. The lamp or LED circuit **must** be functional to enable the ICON™ system.

The above circuit diagram can vary, such as connector or pins, depending on the vehicle make or model. OEM installations can possibly provide the harnessing between the idle control module and other ICON™ devices.

Component Location

The ICON™ lamp or LED is typically located in the vehicle cab on the dash panel.

The ICON™ module can be located in a different location depending on the vehicle application.

Shoptalk

This fault indicates a short circuit to battery voltage.

The ICON™ system can display **only** the present active fault code. If more than one fault code is active at the same time, the ICON™ system flashes out the highest priority fault. After the fault has been corrected then the next active fault will be flashed out.

Note: The ICON™ electronic service tool can display more than one active and or inactive fault codes at the same time.

Warnings and Cautions

⚠ CAUTION ⚠

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement:
Part Number 3822758 - male Deutsch/AMP/Metri-Pack test lead
Part Number 3822917 - female Deutsch/AMP/Metri-Pack test lead.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Read the fault codes.	
	STEP 1A. Use the fault flashout feature or the ICON™ electronic service tool to read the fault codes.	Fault Code 198 inactive
STEP 2.	Check the ICON™ lamp.	
	STEP 2A. Check the ICON™ lamp connector for damaged pins.	No damaged pins

STEPS		SPECIFICATIONS
	STEP 2B. Check for a short circuit to battery.	Less than 0.5 VDC
STEP 3.	Check the ICON™ harnesses.	
	STEP 3A. Inspect the ICON™ engine harness, cab harness, and ICON™ idle control module connector pins.	No damaged pins
	STEP 3B. Check the complete harness for a short circuit from pin to pin.	More than 100k ohms
	STEP 3B-1. Identify whether the ICON™ system is an Aftermarket or an OEM.	ICON™ system is Aftermarket
	STEP 3B-2. Check the cab harness for a short circuit from pin to pin.	More than 100k ohms
	STEP 3C. Check the complete harness for a short circuit to the battery.	Less than 0.5 VDC
	STEP 3C-1. Identify whether the ICON™ system is an Aftermarket or an OEM.	ICON™ system is Aftermarket
	STEP 3C-2. Check the cab harness for a short circuit to the battery.	Less than 0.5 VDC
STEP 4.	Clear the fault code.	
	STEP 4A. Disable the fault code.	Fault Code 198 cleared

STEP 1. Read the fault codes.

STEP 1A. Use the fault flashout feature or the ICON™ electronic service tool to read the fault codes.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect the ICON™ electronic service tool.

Action	Specification/Repair	Next Step

	Fault Code 198 inactive. Refer to Inactive or Intermittent Fault Codes, Procedure 019-362 (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html).	**4A**
		2A

STEP 2. Check the ICON™ lamp.

STEP 2A. Check the ICON™ lamp connector for damaged pins.

Conditions :

- Turn keyswitch OFF.
- Connect the ICON™ electronic service tool to confirm the fault status.
- Disconnect the ICON™ lamp connector from the cab harness.

Action	Specification/Repair	Next Step

- Corroded pins - Bent or broken pins - Pushed back or expanded pins - Wire insulation damage - Moisture in or on the connector - Missing or damaged connector seals - Connector shell broken - Dirt or debris in or on the connector pins. For general inspection techniques, refer to Component Connector and Pin Inspection, Procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	No damaged pins	**2B**
	Repair the damaged pins. - Flush the dirt, debris, or moisture from the connector pins using electrical contact cleaner, Part Number 3824510. - Install the appropriate connector seal if it is damaged or missing. - Repair the lamp connector pins. Refer to Procedures 019-202 (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html) or 019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html). - Repair or replace the OEM wiring harness as necessary.	**4A**

STEP 2B. Check for a short circuit to battery.

Conditions :

- Turn keyswitch OFF.
- Connect the ICON™ electronic service tool to confirm the fault status.
- Disconnect the ICON™ lamp connector from the cab harness.

Action	**Specification/Repair**	**Next Step**

• Measure the voltage from pin A of the ICON™ lamp connector pins to engine block ground. • Measure the voltage from pin B of the ICON™ lamp connector pins to engine block ground. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For multimeter usage techniques, refer to Multimeter Usage, Procedure 019-359 (/qs3/pubsys2/xml/en/procedures/99/99-019-359.html).	Less than 0.5 VDC	**3A**
	Repair or replace the ICON™ lamp assembly. Refer to Procedure 019-046 (/qs3/pubsys2/xml/en/procedures/99/99-019-046.html).	**4A**

STEP 3. Check the ICON™ harnesses.

STEP 3A. Inspect the ICON™ engine harness, cab harness, and ICON™ idle control module connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the ICON™ idle control module A and B connectors from the ICON™ idle control module.
- Disconnect all connectors between the lamp and ICON™ module.

Action	Specification/Repair	Next Step

	No damaged pins	**3B**
• Corroded pins • Bent or broken pins • Pushed back or expanded pins • Wire insulation damage • Moisture in or on the connector • Missing or damaged connector seals • Connector shell broken • Dirt or debris in or on the connector pins. For general inspection techniques, refer to Component Connector and Pin Inspection, Procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	Repair the damaged pins. • Flush the dirt, debris, or moisture from the connector pins using electrical contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. • Repair the ICON™ engine harness. Refer to Procedure 019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html) or 019-207 (/qs3/pubsys2/xml/en/procedures/99/99-019-207.html). • Replace the ICON™ engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html). • Repair the cab harness. Refer to Procedure 019-207 (/qs3/pubsys2/xml/en/procedures/99/99-019-207.html). • Replace the cab harness. Refer to Procedure 019-305 (/qs3/pubsys2/xml/en/procedures/97/97-019-305.html). • Repair or replace the OEM wiring harness as necessary.	**4A**

STEP 3B. Check the complete harness for a short circuit from pin to pin.

Conditions :

- Turn keyswitch OFF.
- Disconnect the ICON™ idle control module A and B connectors from the ICON™ idle control module.
- Disconnect the ICON™ lamp connector from the cab harness.

Action	Specification/Repair	Next Step
• Measure the resistance from pin 5 in the ICON™ idle control module A harness connector to all other pins in the connector. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 100k ohms	**3C**
		3B-1

STEP 3B-1. Identify whether the ICON™ system is an Aftermarket or an OEM.

Conditions :

None

Action	Specification/Repair	Next Step

Refer to Procedure 209-017 (/qs3/pubsys2/xml/en/procedures/97/97-209-017.html).	ICON™ system is Aftermarket	**3B-2**
	Check the OEM wiring harness pin to pin for a short in the lamp output circuit.	**4A**

STEP 3B-2. Check the cab harness for a short circuit from pin to pin.

Conditions :

- Disconnect the 14-pin connector.
- Disconnect the ICON™ lamp or LED.

Action	**Specification/Repair**	**Next Step**

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- Measure the resistance from pin J in the 14-pin pass-through connector, cab harness side, to all other pins except pin K in the connector, cab harness side.

Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification.

For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).

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More than 100k ohms

Repair or replace the ICON™ engine harness.

- Repair the ICON™ engine harness. Refer to Procedure 019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html) or 019-207 (/qs3/pubsys2/xml/en/procedures/99/99-019-207.html).
- Replace the ICON™ engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).

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Repair or replace the ICON™ cab harness

- Repair the cab harness. Refer to Procedure 019-207 (/qs3/pubsys2/xml/en/procedures/99/99-019-207.html).
- Replace the cab harness. Refer to Procedure 019-305 (/qs3/pubsys2/xml/en/procedures/97/97-019-305.html).

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STEP 3C. Check the complete harness for a short circuit to the battery.

Conditions :

- Disconnect the ICON™ idle control module A and B connectors from the ICON™ idle control module. **NOTE:** All other components **must** be connected.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step

- Measure the voltage from pin 5 of the ICON™ idle control module A harness connector to engine block ground. - Measure the voltage from pin 6 of the ICON™ idle control module A harness connector to engine block ground. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For multimeter usage techniques, refer to Multimeter Usage, Procedure 019-359 (/qs3/pubsys2/xml/en/procedures/99/99-019-359.html).	Less than 0.5 VDC Replace the ICON™ idle control module. Refer to Procedure 019-358 (/qs3/pubsys2/xml/en/procedures/97/97-019-358.html).	**Repair Complete**
		3C-1

STEP 3C-1. Identify whether the ICON™ system is an Aftermarket or an OEM.

Conditions :
None

Action	Specification/Repair	Next Step
Refer to Procedure 209-017 (/qs3/pubsys2/xml/en/procedures/97/97-209-017.html).	ICON™ system is Aftermarket	**3C-2**
	Check the OEM wiring harness for a short circuit to battery in the lamp output circuit.	**4A**

STEP 3C-2. Check the cab harness for a short circuit to the battery.

Conditions :

- Disconnect the 14-pin connector. **NOTE:** All other components **must** be connected.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
• Measure the voltage from pin J in the 14-pin pass-through connector, cab harness side, to engine block ground. • Measure the voltage from pin D in the 14-pin pass-through connector, cab harness side, to engine block ground. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For multimeter usage techniques, refer to Multimeter Usage, Procedure 019-359 (/qs3/pubsys2/xml/en/procedures/99/99-019-359.html).	Less than 0.5 VDC Repair or replace the ICON™ engine harness. • Repair the ICON™ engine harness. Refer to Procedure 019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html) or 019-207 (/qs3/pubsys2/xml/en/procedures/99/99-019-207.html). • Replace the ICON™ engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).	**4A**
	Repair or replace the ICON™ cab harness • Repair the cab harness. Refer to Procedure 019-207 (/qs3/pubsys2/xml/en/procedures/99/99-019-207.html). • Replace the cab harness. Refer to Procedure 019-305 (/qs3/pubsys2/xml/en/procedures/97/97-019-305.html).	**4A**

STEP 4. Clear the fault code.

STEP 4A. Disable the fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
• Cycle the keyswitch to verify the fault code is inactive.	Fault Code 198 cleared	**Repair complete**
	Troubleshoot any remaining active fault codes.	**Appropriate troubleshooting charts**

Last Modified: 04-Oct-2004